

Assignment - 8

Pappula Kartikeya Raj
2023001492

Abstract. To understand Propositional Logic in Artificial Intelligence and implement example problems using KANREN, SYMPY and pyDatalog packages in Python.

I. DEFINITION OF PROPOSITIONAL LOGIC IN AI

Propositional Logic in Artificial Intelligence is a form of logic used to represent knowledge using simple statements that are either true or false. These statements are called propositions.

A proposition represents a fact such as:

- It is raining
- The light is on
- The student passed the exam

Propositions can be combined using logical operators such as

- AND ($\wedge$)
- OR ($\vee$)
- NOT ($\neg$)
- IMPLIES ($\rightarrow$)

Propositional logic helps AI systems represent knowledge, perform reasoning and make decisions based on logical rules.

II. PACKAGES USED

A. Kanren

Used for logic programming and relational reasoning.

B. Sympy

Used for symbolic mathematics and logical expressions.

C. pyDatalog

Used for declarative logic programming using facts and rules.

III. STEPS INVOLVED IN PROPOSITIONAL LOGIC

Step 1

Define propositions representing statements.

Step 2

Apply logical operators such as AND, OR, NOT or IMPLICATION.

Step 3

Create logical expressions using propositions.

Step 4

Evaluate the logical expression to determine truth values.

IV. EXAMPLES

A. Example 1

Logical Reasoning using SYMPY

Problem Statement

If it rains then the ground becomes wet.

Let

R = It is raining

W = Ground is wet

Logical rule

$R \rightarrow W$

Step 1

Define logical symbols.

Step 2

Create implication expression.

Step 3

Substitute truth values and evaluate the result.

Program

```
from sympy import symbols
from sympy.logic.boolalg import Implies

R, W = symbols('R W')

expr = Implies(R, W)

print("Logical Expression:", expr)

print("Case 1 R=True W=True:",
      expr.subs({R: True, W: True}))

print("Case 2 R=True W=False:",
      expr.subs({R: True, W: False}))
```

Expected Output

Logical Expression: $R \Rightarrow W$

Case 1 $R=\text{True}$ $W=\text{True}$: True

Case 2 $R=\text{True}$ $W=\text{False}$: False

Explanation

The implication rule states that if R is true then W must also be true. If it is raining but the ground is not wet, the logical expression becomes false.

B. Example 2

Student Pass Condition using KANREN

Problem Statement

If a student studies then the student passes the exam.

Step 1

Define relation for students who study.

Step 2

Add facts about students.

Step 3

Query whether a student studies.

Program

```
from kanren import Relation, facts, run, var

studies = Relation()

facts(studies,
      ("Ravi", True),
      ("Kiran", False))

x = var()

print("Students who study:",
      run(0, x, studies(x, True)))
```

Expected Output

Students who study:
Ravi

Explanation

The system stores facts about students who study.
When queried, the logic system returns the students who satisfy the condition.

C. Example 3

Student Eligibility using pyDatalog

Step 1

Define facts for marks.

Step 2

Define rule for passing.

Step 3

Query students who passed.

Program

```
from pyDatalog import pyDatalog

pyDatalog.clear()
pyDatalog.create_terms('marks, pass_student, X, M')

+ marks("Ravi", 80)
+ marks("Kiran", 40)

pass_student(X) <= (marks(X, M) & (M >= 50))

print("Students who passed:")
print(pass_student(X))
```

Expected Output

Students who passed:
X = Ravi

Explanation

The rule checks whether a student's marks are greater than or equal to 50.
The inference system derives that Ravi passed but Kiran did not pass.

V. RESULT

Propositional logic was successfully studied and implemented using

1. KANREN for logic programming
2. SYMPY for symbolic logical reasoning
3. pyDatalog for rule-based logic reasoning

The examples demonstrate how AI systems evaluate logical statements and derive results using propositional logic.

VI. CONCLUSION

Propositional logic is an important concept in Artificial Intelligence.
It helps represent knowledge using logical statements and enables machines to perform reasoning.

Using Python packages such as KANREN, SYMPY and pyDatalog, propositional logic can be easily implemented to build simple reasoning systems.

REFERENCES

Russell, S. and Norvig, P., Artificial Intelligence A Modern Approach, 3rd Edition, Pearson Education.

SymPy Documentation

<https://docs.sympy.org>

KANREN Python Library

<https://github.com/pythological/kanren>

pyDatalog Documentation

<https://sites.google.com/site/pydatalog/>

Python Documentation

<https://docs.python.org>